

Printed Circuit Board Printer

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Table of Contents

Sr. No.	Topic
1.	Introduction
2.	Design of System • Extrusion System • Linear Motion System
3.	Software for PCB Printer
4.	Week Wise Workflow
5.	Conclusion

Introduction

Traditionally, PCBs are made through the process of subtractive fabrication which etches copper from a board and leaves behind the trace required. 3D-printed PCBs are circuit boards created using additive manufacturing rather than traditional subtractive PCB fabrication. Specialized printers deposit **conductive and insulating materials layer by layer** to build the board.

Conventional PCB Manufacturing (Etching Method)

The most widely used method involves:

- Coating a copper-clad board with a resist layer
- Printing the circuit pattern using UV exposure or toner transfer
- Chemically **etching away unwanted copper**
- Leaving behind the desired conductive traces

Additive PCB Printing (Our Approach)

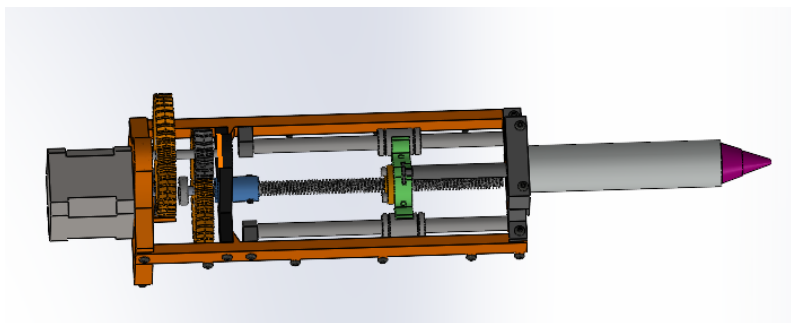
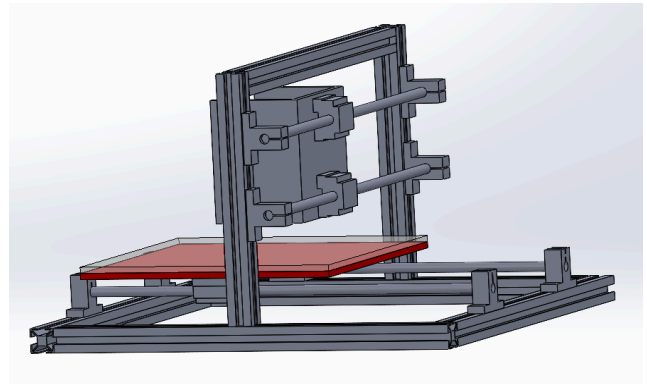
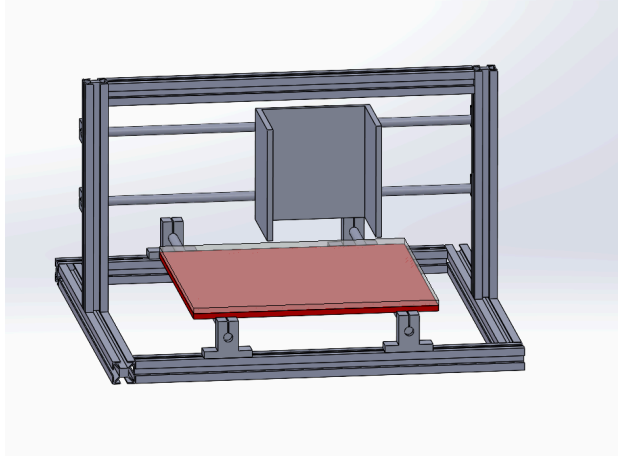
Instead of removing material, conductive ink is **directly deposited** onto a substrate. This is known as **Direct Ink Writing (DIW)**, 3D-printing technique where a printer **extrudes liquid or paste-like material through a nozzle to “draw” structures layer by layer**. These types of PCB 3D printers lay down a conductive material to form the circuitry. These materials usually come as inks or filaments infused with conductive particles, such as silver, copper, or graphite.

- Faster prototyping
- Reduced waste
- Environmentally safer
- Suitable for flexible and custom PCBs

Design of System

Overall, we finalised our design into the following components:

- **Syringe + plunger-based extrusion system** for ink deposition
- **Belt-driven X-Y motion system** for positioning
- **Graphite based ink** to use as our conductive ink
- **Heating bed** to cure the ink



Extrusion System

We explored a variety of extrusion systems for our project including:

1. Filament based extrusion of conductive filament
2. Pneumatic Extrusion system(uses compressed air to force ink through a nozzle for deposition)
3. Screw Driven extruder system(A screw-driven extrusion system uses a rotating screw mechanism to continuously transport ink toward the nozzle)
4. **Piston based extruder system**

A **piston-based extrusion system** uses a mechanically driven plunger, typically powered by a stepper motor and lead screw mechanism, to push ink through the nozzle.

This method provides highly accurate control over extrusion volume and flow rate, making it well-suited for applications requiring precise material deposition. It offers excellent repeatability and is capable of handling medium- to high-viscosity conductive inks commonly used in PCB fabrication.

For the design, we referred to the following research paper: Pandya, K.S., Shindalkar, S.S. & Kandasubramanian, B. Breakthrough to the pragmatic evolution of direct ink writing:

progression, challenges, and future. *Prog Addit Manuf* **8**, 1303–1328 (2023).
<https://doi.org/10.1007/s40964-023-00399-7>

The paper describes the procedure of DIW(Direct Ink Writing), covering the mechanisms and types of ink used for it.

Linear Motion System

Types of Linear Motion Mechanisms

There are various types of systems we could have adopted for our motion in x-y axis. For eg:

1. Lead Screw Drive
2. Rack and Pinon Drive System
3. Linear Rail/Actuator System
- 4. Belt Drive System**

After evaluating the available linear motion systems, the **belt drive system** was selected for the X-Y motion of the 3D printer.

This decision was made because it offers the most practical balance between speed, accuracy, cost, and ease of implementation. The lightweight design enables fast and smooth movement, which is essential for efficient ink deposition and precise path following during PCB printing.

Compared to lead screw systems, the belt drive provides significantly higher movement speed, accuracy. Lead Screw systems are mostly used for Z-axis where there is a lot of load handling. It is also simpler and more economical than rack-and-pinion or advanced linear actuator systems.

Belt Drive Mechanism

Components

- Stepper motors
- Timing belts
- Pulleys
- Linear rails/rods

Working Principle - As the stepper motor rotates, the pulley moves which is connected to the belt. The belt converts rotational motion to translational motion, and the print head moves along the axis.

Software for PCB Printing System

The printer operates in a **Cartesian coordinate system**:

- X-axis: horizontal movement
- Y-axis: forward/backward movement

The specific coordinates of the printhead at any moment are specified by using G-codes. The controller interprets the commands and the stepper motors rotate accordingly. GRBL is an open-source firmware for CNC machines. It runs on microcontrollers (like Arduino) and converts G-code into motor movements.

Belts move the print head precisely, and ink is deposited at the required position.

Overall, it works as follows:

1. PCB design is created in software
2. Design converted into **G-code**
3. G-code sent to GRBL controller
4. GRBL drives stepper motors
5. X-Y system positions the syringe
6. Plunger extrudes ink to form traces

Understanding Gerber Files

The foundation of any PCB project is the *Gerber file* format (RS-274X). These are open-source vector files that describe each physical layer of the board. In this project, the primary file used is the Top Copper (.GTL). While traditionally used to tell a manufacturer where to leave copper on a board, in this context, the Gerber serves as a coordinate map for where conductive material must be deposited.

Traditional Milling vs. Additive Printing

The transition from subtractive to additive manufacturing requires a fundamental shift in toolpath logic:

Subtractive (Milling): The software calculates **Isolation Routing**. The "tool" is a physical bit that removes material around the intended traces. The G-code focuses on the empty space.

Additive (Printing): The software uses **Area Filling (or "Painting")**. The "tool" is a **dispensing nozzle** that deposits material **inside the trace boundaries**. The G-code must account for material flow, volume, and overlap to ensure a solid, conductive path.

The Syringe-Driven Extrusion Setup

To handle high-viscosity conductive ink, the system utilizes a piston-driven extrusion mechanism controlled by a dedicated stepper motor (Z-axis). This mechanical setup necessitates specific modifications to standard CNC G-code:

- **Fluid Dynamics & Pressure:** Unlike a spinning spindle, a syringe has "pressure lag." G-code must include **Priming Sequences** (advancing the Z-axis before X/Y movement) and **Retraction Sequences** (reversing the Z-axis at the end of a trace) to prevent oozing and ensure crisp trace edges.
- **Coordinated Motion:** The Z-axis is synchronized with X and Y. The extrusion rate is calculated as a ratio of the piston's displacement relative to the nozzle's travel distance, ensuring a consistent volume of ink is deposited regardless of movement speed.

Overall, the **Workflow** can be summarised as follows:

1. **Design:** Export Gerber (.GTL) from CAD software.
2. **Path Generation:** Use FlatCAM to generate "Paint Area" geometry based on nozzle diameter.
3. **Conversion:** Transform standard milling commands (M3/M5) into Z-axis extrusion and retraction commands.
4. **Fabrication:** Execute G-code on a GRBL-based CNC Shield, where the Z-axis drives the syringe plunger to deposit conductive traces.

Week Wise Workflow

Week 1: Assigned the project of a 2-D printer. Research on different types of 2-D printers, updated our project to PCB Printer.

Week 2: Researched on different types of linear motions, the working behind the motors. Finalised on the belt drive.

Week 3: Explored different types of extrusion systems, inks, curing methods for the inks.

Week 4: Started with CAD Model for the final project.

Week 5: Tested the motors, linear motion system

Week 6: Started making a prototype of the final extrusion system.

Week 7: Overall assembly of project.

Difficulties faced

- **Non-working stepper drivers and motors:** Multiple stepper motors and drivers that were provided did not work, after trying out multiple combinations, all 3 axes finally worked together. However the setup was interfered with afterwards by someone not part of the team and it led to a non working y axis during final submission.
- **Loose belts:** The belt drive mechanism suffered from belts that kept constantly loosening due to the heavy loads they were carrying. To compensate, a ballscrew

mechanism could be used, however all the available ballscrews were already in use by others.

- **Ink Viscosity:** The extruder mechanism was designed keeping in mind the high viscosity (23,500 cP) as informed by the ink manufacturer. For reference, water has a viscosity of 1 cP. However, the ink that arrived was far less viscous and resembled water more than viscous fluids. By then, it was too late to change the ink deposition mechanism.

Conclusion

The syringe-based extrusion PCB printer represents a shift from traditional subtractive manufacturing to a more efficient **additive approach**. By combining precise ink deposition with a belt-driven motion system and GRBL control, the system achieves a balance between **cost, performance, and flexibility**, making it highly suitable for rapid prototyping and experimental electronics.

The technology is still new and being researched with only a few commercially available PCB printers, which are very expensive and use their own developed inks. Hence, the project was an attempt to make an innovative prototype PCB printer which could be worked upon further and perfected to develop a low cost printer on the commercial market.